

Understanding Reliability In Bluetooth Technology

This article discusses Bluetooth reliability in an in-depth manner and addresses the many connectivity obstacles for optimal use of this wireless communication technology. Effective data transmission and reception requires a careful grasp of packet delivery over different instances of time.

Like other wireless communication technologies, Bluetooth also uses radio. But radio as a medium for encoding and conveying data is not reliable as there are all sorts of challenges present. So not just Bluetooth, any communication technology that is built on radio at the physical layer has to contend with a whole range of fundamental problems. But with the right stack design and the right processes in the various layers, a highly reliable radio communication can be established.

In simple words, a reliable Bluetooth communication takes place when the data transmitted equals the data received and any intended action relating to that data happens as expected. In the real world, however, there can sometimes be more nuanced definitions, subject to different reliability parameters involved. For example:

Tolerance for failure: A 99.9999% success rate may be deemed reliable.

Latency: A low system delay is required for intended actions to be observed within 500 milliseconds of data transmission.

Transactions: Series of updates where either all changes are applied successfully or none of the changes are made.

Resilience: The system successfully processes operations even after certain components face failure/deterioration.

Challenges to radio based technologies

Radio communications take place within channels of the radio spectrum. If two devices transmit on the same channel at the same time, they corrupt each other—resulting in data loss. This is known as collision. The probability of its occurrence is determined by the recurrence and time taken for

transmission. Adding to that is the transmission speed and packet size.

As you all know, Bluetooth uses the 2.4GHz band, which is also known as ISM. Wi-Fi, ZigBee, and DECT phones can use this band as well.

Radios in Bluetooth devices are generally half-duplex, that is, they can both transmit and receive but not at the same time. Take for instance a scenario where a packet is transmitted on a channel by device A to device B.

For the reception, device B must be actively listening on the correct channel at the same time. If it is not listening, or is on a different channel, the packets will not be received. The proportion of time a device spends listening is called the Rx duty cycle. Do not forget, a device can listen only on a single channel at a time.

Signal strength also has a huge impact. One is likely to experience more errors when signal strength is low (due to it being relatively close to background noise). The Bluetooth core specification states that a bit error rate (BER) of up to 0.1% is permitted. Anything beyond that is not acceptable.

The signal strength at which a radio can stay within this limit is known as its receiver sensitivity. So, if you are a product designer, choosing a radio with good receiver sensitivity is important.

Stack implementations often include several buffers for temporary packet storage. If the rate of packet arrival into a buffer exceeds the rate of packet consumption for a long enough period

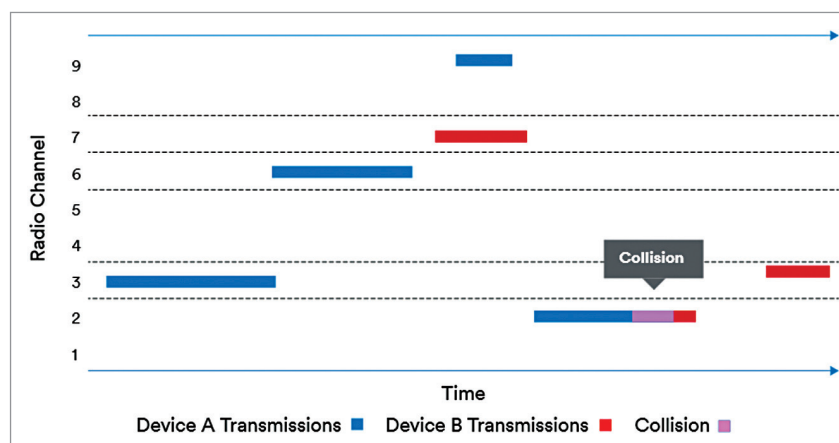


Figure 1: Collision taking place on channel number 2